Lecture 7 - Intro to Prior Distributions and Single Parameter Bayesian Models

DS 4420: Machine Learning and Data Mining 2

Eric Gerber
e.gerber@northeastern.edu
Northeastern University, Boston, USA

Northeastern University

Today

1. **Bayes' Rule**
2. **Prior Distributions**
3. **Single Parameter Models**
4. **In Python and R**
5. **Next Steps**

Today

1. Bayes' Rule
2. Prior Distributions
3. Single Parameter Models
4. In Python and R
5. Next Steps

Updates

- Homework 5 will be assigned **Monday, Mar. 9**
  - Will be due by **Monday, Mar. 23 at 11:59 pm** (by **Sunday, Mar. 22 at 11:59 pm** for 5% extra credit)
- Project Phase I due date pushed to **Friday, Mar. 13 by 11:59 pm**
- Plan is to have Quiz 5 either **Wednesday or Thursday, Mar. 11 or 12** (on this material: it will depend on how quickly we get through this lesson)

Topics for Today:

1. Revisit Bayes' Rule
2. Prior Distributions
3. Single Parameter Models

Push-Up Counter:

- Section 1: 13
- Section 4: 9

Today

1. **Bayes' Rule**
2. Prior Distributions
3. Single Parameter Models
4. In Python and R
5. Next Steps

Bayes' Rule

Bayes' Rule is simply a rearranging of the conditional probability formula:

$$p(x|y) = \frac{p(x,y)}{p(y)} = \frac{p(y|x)p(x)}{p(y)}$$

- The Bayes Rule forms the basis for many statistical supervised machine learning models
- We will be spending a long time with this the rest of the semester!

Thomas Bayes

Bayes' Rule

With simple (naive) Bayesian problems, we are used to having exact numbers. For example, in a neighborhood of sick children (assume all the children are sick...):

- 90% of children are getting the flu $p(f) = .9$
- 10% are getting measles $p(m) = .1$
- The probability a child has a rash given they have flu is $p(r|f) = .08$
- The probability a child has a rash given they have measles is $p(r|m) = .95$

What is the probability a child has the flu, given they have a rash, $p(f|r)$?

Easy!

$$p(f|r) = \frac{p(r|f)p(f)}{p(r)} = \frac{p(r|f)p(f)}{p(r|f)p(f) + p(r|m)p(m)} = \frac{.08 \times .9}{.08 \times .9 + .95 \times .1} \approx .431$$

BUT, in real life, we don't know these numbers; we will only assume we know the **distributions** they come from.

Motivating Example

Suppose you want to estimate the probability your crush is interested in you based on if they text you back or not. This is a binary problem, perhaps most simple to assume it follows a Bernoulli distribution:

$$p(y|\theta) = \theta^y(1-\theta)^{1-y}$$

Where $y = 1$ if your crush texts you back and $y = 0$ if they don't, and θ is the probability they are interested in you.

Not So Easy: What is θ?

How can we make use of Bayes' rule to give us an idea of what θ is?

Anatomy of Bayes Rule

Each component of Bayes' rule has it's own name worth discussing:

$$p(\theta|y) = \frac{p(y|\theta) \times p(\theta)}{p(y)}$$

1. The **Prior**: $p(\theta)$ is the "prior knowledge" about the parameter in this situation
   - In our example, perhaps you believe (from "prior" experience) that 50% of everyone in the world is interested in you.
2. The **Likelihood**: $p(y|\theta)$ is the likelihood of an event given your prior knowledge
   - Here, it would be the probability of your crush texts you back, assuming you know θ.
3. The **Evidence/Marginal:** $p(y)$ is the observation of what actually happened.
   - You can observe if they text you back or not; we are assuming that gives us some information about if they are interested or not, so serves as our basis.
4. The **Posterior**: $p(\theta|y)$ is the ultimate question we are trying to answer.
   - Given the evidence/if your crush texts you back, what is the probability they are interested in you?

Motivating Example

In our motivating example, we have a very reasonable distribution for the Likelihood:

$$p(y|\theta) = \theta^y(1-\theta)^{1-y}$$

But to find the Posterior Distribution we still need to determine what the Prior on θ is, and then **marginalize** in order to find the Evidence/Marginal.

Think Critically

What might be a reasonable distribution for θ? What do we know about it?

- θ is a probability (between 0 and 1)
- We need some prior "belief" about how θ behaves to choose $p(\theta)$
- Technically, $p(\theta)$ could be anything...

Today

Picking Priors

What if we picked an exponential distribution for θ? Recall we said, as an example, that we believe 50% of the world is interested in you. This implies we believe $E(\theta) = .5$. If you recall, the exponential distribution has the pdf:

$$p(x) = \lambda e^{-\lambda x}$$

- where $E(X) = \frac{1}{\lambda}$

This would imply we believe the distribution of θ to be:

$$p(\theta) = 2e^{-2\theta}$$

Picking Priors

Given the exponential prior, we can then find the **joint** distribution (or, the numerator of the Posterior), because we have both the Bernoulli likelihood and the exponential prior, so:

$$p(y,\theta) = p(y|\theta)p(\theta) = \left[\theta^{y}(1-\theta)^{1-y}\right]\left[2e^{-2\theta}\right]$$

Picking Priors

In Bayesian ML, we can normally collect the data to model the Likelihood, and can technically choose any distribution for the Prior, however in order to solve for the Posterior, note a couple things (using our specific example):

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)} = \frac{p(y|\theta)p(\theta)}{\int_\theta p(y|\theta)p(\theta)d\theta} = \frac{\left[\theta^y(1-\theta)^{1-y}\right]\left[2e^{-2\theta}\right]}{\int_0^\infty \left[\theta^y(1-\theta)^{1-y}\right]\left[2e^{-2\theta}\right]d\theta}$$

1. Is the exponential even appropriate here? If θ can only be between 0 and 1, why specify a prior distribution that could be infinite?
2. Even if (in this case) the Marginal integral isn't too complicated, it is definitely the most difficult aspect of this calculation!
   - For some problems, it is **impossible** to solve directly...

Conjugate Priors

Let's investigate a more reasonable prior distribution for the Bernoulli. We previously talked about the Beta distribution, which can be between 0 and 1:

$$p(\theta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha,\beta)}$$

- where α and β are the parameters of the distribution (note that $\alpha = \beta = 1$ is equivalent to the Uniform from 0 to 1)
- and where (important later) $B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$

What makes this a better prior distribution? It is what we call a Conjugate Prior.

Conjugate Priors

A Conjugate Prior is a prior distribution chosen so that the Posterior distribution is the **same as the prior distribution**. Let's prove this is the case for the Bernoulli Likelihood with Beta Prior over the next few slides. Recall the set up:

$$y|\theta \sim Bern(\theta), \quad y \in \{0, 1\}$$

$$\theta \sim Beta(\alpha, \beta)$$

Goal: Find $p(\theta|y)$

The Beta as a Conjugate Prior for the Bernoulli

Likelihood

$p(y|\theta) = \theta^y (1-\theta)^{1-y}$

Prior

$p(\theta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha,\beta)}$

Joint

$$p(y,\theta) = \left[\theta^y (1-\theta)^{1-y}\right] \left[\frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha,\beta)}\right]$$

$$= \frac{1}{B(\alpha,\beta)} \theta^{y+\alpha-1} (1-\theta)^{1-y+\beta-1}$$

- Note: I'm not simplifying the last exponent for a reason...

The Beta as a Conjugate Prior for the Bernoulli

Marginal

$$\begin{aligned} p(y) &= \int_0^1 \frac{1}{B(\alpha,\beta)} \theta^{y+\alpha-1}(1-\theta)^{1-y+\beta-1} d\theta \\ &= \frac{1}{B(\alpha,\beta)} \left[B(y+\alpha, 1-y+\beta)\right] \end{aligned}$$

- Note: Remember the definition of $B(x,y)$!

Posterior

$$\begin{aligned} p(\theta|y) &= \frac{\frac{1}{B(\alpha,\beta)} \times \theta^{y+\alpha-1}(1-\theta)^{1-y+\beta-1}}{\frac{1}{B(\alpha,\beta)} \times B(y+\alpha, 1-y+\beta)} \\ &= \frac{\theta^{y+\alpha-1}(1-\theta)^{1-y+\beta-1}}{B(y+\alpha, 1-y+\beta)} \end{aligned}$$

Which, if you are a little clever/have been paying attention, means the posterior, $\theta|y \sim Beta(y+\alpha, 1-y+\beta)$

Putting It Together

Suppose we believe that the probability someone is actually interested in us is 5%, but we would like to collect some data to update this. Assuming a Beta prior distribution, we can pick α and β so that $E(\theta) = .05$. One option is: $\alpha = 1, \beta = 19$, then: $E(\theta) = \frac{\alpha}{\alpha+\beta} = \frac{1}{20} = .05$.

- You text the person.
- They text you back!
- How is your belief in if the person is interested updated? (i.e. what is the mean of the posterior distribution?)

Putting It Together

If you were to use the MLE of the data you collected to inform your belief, with only a single sample (they texted you back!) you would say, well $\frac{1}{1} = 1$, so 100%, they are interested!

- A Bayesian, even if they are relying on "belief", is more realistic!
- Because the posterior distribution is $\theta|y \sim Beta(\alpha^* = y + \alpha, \beta^* = 1 - y + \beta)$, a Bayesian would say their updated estimate of the chances someone is interested is:

$$E(\theta|y) = \frac{\alpha^*}{\alpha^* + \beta^*} = \frac{y + \alpha}{y + \alpha + 1 - y + \beta} = \frac{1 + 1}{1 + 1 + 1 - 1 + 19} = \frac{2}{21} \approx .095$$

- On the other hand, if your crush didn't text you back, instead of all hope being lost, a Bayesian would tell you: $E(\theta|y = 0) = \frac{1}{21} \approx .048$

Common Conjugate Prior Distributions

Note I will be asking you to prove (the same way we just did it for the Beta-Bernoulli) some of these relationships either on a homework, a quiz, or on the final exam:

Data Likelihood	Conjugate Prior
Bernoulli (Binomial)	Beta
Categorical (Multinomial)	Dirichlet
Gaussian (known variance)	Gaussian
Gaussian (unknown variance)	Normal-Inverse-Gamma
Multivariate Gaussian (known covariance)	Multivariate Gaussian
Multivariate Gaussian (unknown covariance)	Normal-Inverse-Wishart
Isotropic Gaussian (unknown covariance)	Normal-Inverse-Gamma
Poisson	Gamma
Exponential	Gamma

Wikipedia also has a very nice table/list.

- Both the Dirichlet and the Gamma distributions are not too difficult, though they both rely on the Gamma Function, $\Gamma(x) = \int_0^\infty t^{x-1}e^{-t}dt$ or, when x is an integer: $\Gamma(x) = (x-1)!$

Today

Single Parameter Models

These simple models are the foundation of Bayesian analysis, though our life will get quite a bit more complicated soon with multi-parameter models.

Before that, consider what we just learned and explored:

- Given data and an assumed distribution with unknown parameter(s)
- We can use Bayes' rule to estimate the distribution of the parameter(s) given the data and some prior belief about the distribution of the parameter(s)

Single Parameter Models

The use of data is key to the implementation of Bayesian modeling. Notice how with only a single observation in our example (one text or not), the posterior distribution did not change very much from the prior:

$$\theta \sim Beta(\alpha = 1, \beta = 19)$$

$$\theta|y = 1 \sim Beta(\alpha^* = 2, \beta^* = 19)$$

$$\theta|y = 0 \sim Beta(\alpha^* = 1, \beta^* = 20)$$

Essentially, we have very little information/data/evidence ($n = 1$), so the posterior distribution is "driven" more by the prior than the data itself.

- What if we had a lot of data?

Extending the Model

Imagine you now have sent $n = 30$ texts to your crush, and believe that the number they respond to will give you some information on the probability they are interested in you. We can easily show that (if you assume each text is independent), the joint likelihood of n Bernoulli random variables also has the Beta as a conjugate prior. Do everything again, but now with:

$$p(y|\theta) = \prod_{i=1}^{30} p(y_i) = \prod_{i=1}^{30} \theta^{y_i}(1-\theta)^{1-y_i}$$

Or, to set up again:

$$y_i|\theta \sim Bern(\theta), \quad y_i \in \{0,1\}, \quad i = 1, \cdots, 30$$
$$\theta \sim Beta(\alpha, \beta)$$

Goal: Find $p(\theta|y)$

Do It Again

Likelihood

$p(y|\theta) = \prod_{i=1}^{30} \theta^{y_i}(1-\theta)^{1-y_i} = \theta^{\sum y_i}(1-\theta)^{n-\sum y_i}$

Prior

$p(\theta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha,\beta)}$

Joint

$$p(y,\theta) = \left[\theta^{\sum y_i}(1-\theta)^{n-\sum y_i}\right]\left[\frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha,\beta)}\right]$$

$$= \frac{1}{B(\alpha,\beta)}\theta^{\sum y_i+\alpha-1}(1-\theta)^{n-\sum y_i+\beta-1}$$

Do It Again

Marginal

$$p(y) = \int_0^1 \frac{1}{B(\alpha,\beta)} \theta^{\sum y_i + \alpha - 1}(1-\theta)^{n - \sum y_i + \beta - 1} d\theta$$
$$= \frac{1}{B(\alpha,\beta)} \left[B\left(\sum y_i + \alpha, n - \sum y_i + \beta \right) \right]$$

- Note: Remember the definition of $B(x, y)$!

Posterior

$$p(\theta|y) = \frac{\frac{1}{B(\alpha,\beta)} \theta^{\sum y_i + \alpha - 1}(1-\theta)^{n - \sum y_i + \beta - 1}}{\frac{1}{B(\alpha,\beta)} B\left(\sum y_i + \alpha, n - \sum y_i + \beta\right)}$$
$$= \frac{\theta^{\sum y_i + \alpha - 1}(1-\theta)^{n - \sum y_i + \beta - 1}}{B\left(\sum y_i + \alpha, n - \sum y_i + \beta\right)}$$

Which, if you are a little clever/have been paying attention, means the posterior, $\theta|y \sim Beta(\sum y_i + \alpha, n - \sum y_i + \beta)$

More Data = Better Estimates (Still True in Bayesian Modeling)

Notice that the Beta is still the conjugate prior for a bunch of Bernoulli, but now we can see how the data/evidence impact the posterior relative to the prior:

- The Posterior Distribution is: $\theta|y \sim Beta(\sum y_i + \alpha, n - \sum y_i + \beta)$
  - We were using hyperparameters $\alpha = 1$ and $\beta = 19$
- When $n = 1$, we saw that the posterior mean (our best estimate for the probability your crush was interested) changed very little.
- However, now let's say your crush responds to all 30 of your messages. How would the Bayes' estimator be updated?

$$E(\theta|y) = \frac{30 + 1}{(30 + 1) + (30 - 30 + 19)} = \frac{31}{50} = .62$$

- Whoa! Even though our original belief was that it was that the probability anyone would be interested was 5%, given the many texts that were responded to, now we're much more confident!

More Data = Better Estimates (Still True in Bayesian Modeling)

- This illustrates one part of how the likelihood and prior "compete" to drive the Posterior. With enough data, your hyperparameter choices are less impactful.
- Note also: since α and β are fixed beforehand, based on prior belief, you might use very small values of α and β if you are not confident in the prior distribution:

Non-Informative (and Improper) Priors

A Beta distribution with $\alpha = 0$ and $\beta = 0$ is undefined (this means it would be considered an improper prior), but what would happen to our posterior mean if we used it? Recall that the posterior distribution with it's expected value is:

$$\theta|y \sim Beta(\sum y_i + \alpha, n - \sum y_i + \beta)$$

$$E(\theta|y) = \frac{\sum y_i + \alpha}{(\sum y_i + \alpha) + (n - \sum y_i + \beta)}$$

If you use the completely non-informative (and in this case improper) prior $\theta \sim Beta(\alpha = 0, \beta = 0)$, what does $E(\theta|y)$ turn into?

$$E(\theta|y) = \frac{\sum y_i + 0}{(\sum y_i + 0) + (n - \sum y_i + 0)} = \frac{\sum y_i}{n} = \overline{y}$$

The maximum likelihood estimator! (As if you never used Bayes' rule at all...)

Other Distributions You'll Want to Know

Gamma (Use on HW)

$$p(x|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$$

Where $0 \leq x < \infty$, $\alpha > 0$, and $\beta > 0$

- The conjugate prior to the exponential. We briefly talked about it in Lecture 6.

Dirichlet (Use Soon)

$$p(x|\alpha) = \frac{1}{B(\alpha)} \prod_{i=1}^{K} x_i^{\alpha_i - 1}$$

Where $K \geq 2$, $0 \leq x \leq 1$, $\sum_{i=1}^{K} x_i = 1$, $\alpha_i > 0$, $B(\alpha) = \frac{\prod_{i=1}^{K} \Gamma(\alpha_i)}{\Gamma(\alpha_0)}$, and $\alpha_0 = \sum_{i=1}^{K} \alpha_i$.

- The conjugate prior for the categorical/multinomial. It's new, but hopefully not too bad.

Another Example

Pareto

$$p(x|\wedge, \alpha) = \frac{\alpha \wedge^{\alpha}}{x^{\alpha+1}}$$

Where $\wedge \leq x < \infty$, thus $\wedge = \min(x) > 0$ and $\alpha > 0$

- Let's show this is a conjugate prior for θ in $Y \sim Unif(0, \theta)$.

Another Example

$$y|\theta \sim Unif(0, \theta), \quad 0 \leq y \leq \theta$$
$$\theta \sim Pareto(\wedge, \alpha)$$

Goal: Find $p(\theta|y)$

Another Example

Likelihood

$p(y|\theta) = \frac{1}{\theta}$

Prior

$p(\theta) = \frac{\alpha \wedge^{\alpha}}{\theta^{\alpha+1}}$

Joint

$$p(y, \theta) = \left[\frac{1}{\theta}\right]\left[\frac{\alpha \wedge^{\alpha}}{\theta^{\alpha+1}}\right]$$
$$= \frac{\alpha \wedge^{\alpha}}{\theta^{\alpha+2}}$$

Practice

Given the distributions on the previous slide, and the below marginal distribution, find the posterior $\theta|y$ and prove that it is also a Pareto distribution, defining the posterior parameters.

Marginal

$$p(y) = \text{...after some math...} = \frac{\alpha\wedge^{\alpha}}{(\alpha+1)\vee^{\alpha+1}}$$

Where $\vee = \max(y, \wedge)$ (the maximum value between the data or prior $\wedge$).

Posterior

$$p(\theta|y) = \cdots$$

Today

1. Bayes' Rule
2. Prior Distributions
3. Single Parameter Models
4. **In Python and R**
5. Next Steps

In Python and R, With Real Data

We can visualize how these distributions (the prior, the posterior, and the likelihood) behave with plots given some real data. Check out the **plotting_bayes** files (in both Python and R) to visualize the Pareto-Uniform distribution in action.

Note

For now, we are assuming we have manually derived (and thus are able to manually derive) the posterior distribution. As we move to more complex models, we may need to use pre-built packages in python and R to help us!

Today

1. Bayes' Rule
2. Prior Distributions
3. Single Parameter Models
4. In Python and R
5. Next Steps

What if the Marginal Is Too Difficult?

Sometimes models have a very difficult to calculate posterior, because the marginalization step is quite difficult and/or annoying. For example, suppose you have the categorical distribution:

$$p(y|\theta) = \prod_{i=1}^{K} \theta_i^{y_i}$$

Proving rigorously that the Dirichlet is the conjugate prior is somewhat rough because of the marginal:

$$p(\theta) = \frac{1}{B(\alpha)} \prod_{i=1}^{K} \theta_i^{\alpha_i - 1}$$

$$p(y, \theta) = \frac{1}{B(\alpha)} \prod_{i=1}^{K} \theta_i^{y_i + \alpha_i - 1}$$

$$p(y) = \int_0^1 \int_0^1 \cdots \int_0^1 \frac{1}{B(\alpha)} \prod_{i=1}^{K} \theta_i^{y_i + \alpha_i - 1} d\theta_1 \cdots d\theta_{K-1} d\theta_K$$

What if the Marginal Is Too Difficult?

The larger K is, the more annoying that integral becomes! However, one shortcut is to essentially "ignore" the evidence/marginal, and say we know that the posterior distribution is proportional to the likelihood times the prior:

$$p(\theta|y) \propto p(y|\theta)p(\theta)$$

When we do that, if we can (sometimes not so easily) identify the essential parts of a distribution from only the numerator, we can say that ("to a normalizing constant") we know the posterior distribution. In this example:

$$p(\theta|y) \propto p(y,\theta) = \frac{1}{B(\alpha)} \prod_{i=1}^{K} \theta_i^{y_i+\alpha_i-1}$$

Note that $\frac{1}{B(\alpha)}$ is a constant w.r.t θ:

$$p(\theta|y) \propto \prod_{i=1}^{K} \theta_i^{y_i+\alpha_i-1}$$

- The trick now is to notice that this looks like everything (except the normalizing constant) we would need for a Dirichlet distribution with parameters $\alpha_i^* = y_i + \alpha_i$ (which is conjugate)
- **Note:** it's not always *that* easy...

More Complex Models

The Bayesian models we discuss here are what we call single parameter models, because we only need a single prior distribution for a single parameter, and we will assume we know the hyperparameters. Moving forward:

1. Simple multi-parameter models:
   - For example: the Categorical (just seen) and the Multivariate Gaussian (We'll deal with this next.)
2. Models with inputs (Bayesian supervised machine learning):
   - When the unknown parameter(s) involve functions of inputs, we now place prior distribution(s) on the weights of those inputs
   - We may or may not deal with these in Bayesian supervised learning.

Bayesian logistic regression:

$$y_i|\theta_i \sim Bern(\theta_i)$$

$$\theta_i = \frac{1}{1 + e^{-w^T x_i}}$$

$$w \sim MVN_d(\mu, \Sigma)$$